

Chapter 7.

$$7.1. \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 1 \\ -3 \end{bmatrix}$$

$$7.2. \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \\ 1 \\ 6 \end{bmatrix}$$

$$7.3. \quad L = \begin{bmatrix} 1 & & & & \\ -\frac{1}{2} & 1 & & & \\ & -\frac{2}{3} & 1 & & \\ & & -\frac{3}{4} & 1 & \\ & & & -\frac{4}{5} & 1 \end{bmatrix} \quad U = \begin{bmatrix} 2 & -1 & & & \\ & \frac{3}{2} & -1 & & \\ & & \frac{4}{3} & -1 & \\ & & & \frac{5}{4} & -1 \\ & & & & \frac{6}{5} \end{bmatrix}$$

General Case:

$$L = \begin{bmatrix} 1 & & & & \\ -\frac{1}{2} & 1 & & & \\ & -\frac{2}{3} & 1 & & \\ & & \ddots & \ddots & \\ & & & -\frac{n-2}{n-1} & 1 \\ & & & & -\frac{n-1}{n} & 1 \end{bmatrix} \quad U = \begin{bmatrix} 2 & -1 & & & \\ & \frac{3}{2} & -1 & & \\ & & \frac{4}{3} & -1 & \\ & & & \ddots & \ddots \\ & & & & \frac{n}{n-1} & -1 \\ & & & & & \frac{n+1}{n} \end{bmatrix}$$

7.4. (a) Yes (b) No (c) Yes.

7.5. Only (c) satisfies the conditions of Theorem 7.8.

7.6. No. for example: $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$.

A satisfies all conditions of Theorem 7.8, but A is even not nonsingular.